

# Generalizable Human Modeling for

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## Physical Human–Robot Co-Manipulation

Connecting multimodal interaction, structured human models,  
and safe robot planning and control

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Research Discussion with RISE Lab

# Co-manipulation requires coordination through human uncertainty

## Physical co-manipulation

A human and a robot physically manipulate a shared object to accomplish a common task safely and efficiently.

### The robot must simultaneously:

- ▶ infer the human's intent and interaction strategy;
- ▶ predict responses to robot actions;
- ▶ coordinate motion and interaction forces;
- ▶ adapt to individual behavior and strategy changes;
- ▶ satisfy physical, actuation, and safety constraints.

## Central research question

How can the robot model the human well enough to support real-time planning and control?

## Desired outcome

An explicit human model that is predictive, interactive, uncertainty-aware, and adaptable online.

# Human behavior is structured, adaptive, and bounded-rational

- ▶ **Multimodal**: multiple strategies may be reasonable in the same situation.
- ▶ **Interactive**: behavior depends on what the robot does.
- ▶ **Bounded-rational**: people may be delayed, inconsistent, or effort-limited.
- ▶ **Adaptive**: people infer the robot's intent and change their behavior.
- ▶ **Person-dependent**: roles, preferences, and control styles differ.
- ▶ **Task-conditioned**: concrete goals change, but some latent concepts transfer.

## Implication

A useful model must distinguish human intent, interaction style, adaptation, and action uncertainty—without requiring a separate model for every task.

# The shared papers provide complementary foundations

## Diffusion Co-Policy

- ▶ Learns multimodal collaborative action sequences
- ▶ Demonstrates mutual adaptation and role switching
- ▶ Works with real humans in collaborative carrying

## Bounded-Rational Model

- ▶ Models human joint action game-theoretically
- ▶ Fits behavior better with bounded rationality
- ▶ Reveals leadership and adaptation effects

## PACE / N-PACE

- ▶ Infers intent and cost parameters online
- ▶ Treats the peer as another adapting learner
- ▶ Connects inference with interaction policies

## Combined insight

Learned models provide behavioral flexibility; game-theoretic models provide structure, interpretability, and a connection to control.

# A human-modeling gap remains in physical interaction

## Learned co-policies

- ▶ Model the human mostly implicitly
- ▶ Often require task-specific demonstrations
- ▶ Do not explicitly represent intent, belief, or preference
- ▶ Offer limited physical and safety guarantees

## Game / IOC models

- ▶ Require predefined cost features
- ▶ Often assume known intent or learning structures
- ▶ Struggle with multimodal, history-dependent behavior
- ▶ May require redesign when the task changes

## Research gap

Physical co-manipulation needs an explicit but flexible human model that connects motion, force/torque, and task context to robot planning and control.

# Predict human responses to candidate robot actions

## Passive prediction

$$p(u_{t:t+H}^H \mid h_t, c)$$

“What will the human do next?”

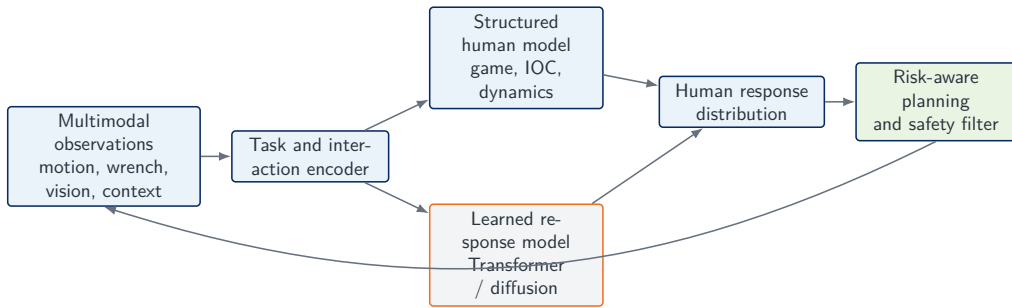
## Interactive prediction

$$p_{\psi}\left(u_{t:t+H}^H, z_{t+1}^H \mid h_t, c, u_{t:t+H}^R\right)$$

“How will the human respond if the robot takes this action?”

- ▶  $h_t$ : human, robot, object, and interaction-wrench history
- ▶  $c$ : task goal, object properties, geometry, contacts, and constraints
- ▶  $u_{t:t+H}^R$ : candidate robot actions;  $z^H$ : latent human state

# A hybrid architecture keeps the human model explicit



## Design principle

The structured component explains behavior; the learned component captures multimodality and model mismatch; online adaptation personalizes both to the current human.

# SP1: Build the physical evidence base

## Theme and focus

### Physically grounded multimodal data

- ▶ Build shared-object transport experiments
- ▶ Synchronize motion, object state, and force/torque
- ▶ Create ambiguous paths, role switching, and strategy changes
- ▶ Compare motion-only, wrench-only, and multimodal inference

## Challenges → solutions

- ▶ No direct intent labels → hierarchical task/strategy/action labels
- ▶ Noisy contact data → filtering and contact-state estimation
- ▶ High person variability → probabilistic latent-state models
- ▶ Costly data → human–human, simulation, and HRI data mixture

**Expected paper:** A multimodal benchmark and systematic study of force/torque for human intent inference.

# SP2: Make intent and preference explicit

## Bounded-rational human model

$$J_H = \sum_t w_t^{H^\top} \phi_\psi(x_t, u_t^H, u_t^R, c)$$

Reusable features may represent progress, pose, effort, interaction force, safety, smoothness, and coordination.

- ▶ Infer intent, role, preference, and rationality online
- ▶ Represent the human's belief about robot intent
- ▶ Treat the human as another adapting learner

## Challenges → solutions

- ▶ Cost identifiability → Bayesian / maximum-entropy IOC
- ▶ Ambiguous intent → informative robot actions
- ▶ Unknown adaptation → latent learning-state estimation
- ▶ Online computation → local game approximations and MPC

**Expected paper:** Peer-aware intent and preference inference for physical co-manipulation.

# SP3: Learn responses to possible robot actions

## Interactive response prediction

$$p_{\psi}(u_{\text{future}}^H \mid h_t, c, u_{\text{future}}^R)$$

- ▶ Predict multimodal human response trajectories
- ▶ Condition on motion, wrench, context, and candidate robot actions
- ▶ Predict uncertainty and latent-state transitions
- ▶ Compare Transformer, diffusion, and latent-variable models

## Challenges → solutions

- ▶ Limited action coverage → diverse robot probing
- ▶ Correlation vs. response → counterfactual conditioning
- ▶ OOD actions → uncertainty estimation and detection
- ▶ Slow inference → distilled or few-step diffusion

**Expected paper:** Robot-action-conditioned diffusion models for multimodal human response prediction.

# SP4: Close the loop through safe control

## Human-model-aware planning

$$u_R^* = \arg \min_{u_R} \mathbb{E}_{u_H \sim p_\psi} [J_R]$$

subject to object dynamics, actuation, force, collision, and safety constraints.

- ▶ Optimize progress, team effort, and interaction force
- ▶ Replan as the human model is updated
- ▶ Use a safety filter before execution

## Challenges → solutions

- ▶ Prediction error → short-horizon MPC
- ▶ Distributional complexity → scenario sampling
- ▶ Excess conservatism → risk-sensitive objectives
- ▶ Hard safety limits → CBF / constrained safety filter

**Expected paper:** Uncertainty-aware planning and safe control with interactive human response models.

# SP5: Transfer human concepts across tasks and people

## Task-conditioned generalization

Shared human latent concepts:

- ▶ intent and local strategy;
- ▶ leader–follower role and compliance;
- ▶ effort, safety, and interaction preferences;
- ▶ reaction delay and adaptation state.

Encode goals, object dynamics, geometry, contacts, and constraints as task context. Adapt only latent states, preference weights, or a small adapter online.

**Expected paper:** A task-conditioned human model for cross-task co-manipulation and online personalization.

## Challenges → solutions

- ▶ Different task structures → object-centric states and wrenches
- ▶ Limited physical data → simulation and human–human pretraining
- ▶ Negative transfer → structured task encoders
- ▶ Unstable personalization → Bayesian updates or small adapters

# Five subprojects answer one connected question chain

**SP1 — Observe:** What physical signals reveal human intent and interaction strategy?



**SP2 — Infer:** What does the human intend, prefer, and believe about the robot?



**SP3 — Predict:** How will the human respond to candidate robot actions?



**SP4 — Act:** How should the robot coordinate safely under prediction uncertainty?



**SP5 — Transfer:** What representations generalize across people and tasks?

# A focused transport study can test the core hypothesis

## Initial task

A human and robot transport a shared object through an environment with multiple feasible paths.

## Model inputs

- ▶ Object and human motion
- ▶ Interaction force/torque
- ▶ Goal and environment geometry
- ▶ Candidate robot actions

## Comparisons

1. Reactive control without a human model
2. Passive prediction from interaction history
3. Structured intent inference
4. Robot-action-conditioned prediction
5. Hybrid structured + learned model

## Evaluation

- ▶ Prediction accuracy and calibration
- ▶ Success, time, and team effort
- ▶ Interaction force and disagreement
- ▶ Adaptation, safety, and comfort violations

# The program connects human understanding to robot action

1. A physically grounded multimodal benchmark
2. Bounded-rational, peer-aware intent inference
3. Interactive prediction of responses to robot actions
4. Safe planning over human response distributions
5. Cross-task generalization and online personalization

## Long-term vision

A versatile human model that connects physical interaction, human adaptation, and robot decision-making across co-manipulation scenarios.

## Proposed starting point

Use a physical collaborative-transport task to test whether force/torque sensing and robot-action-conditioned prediction improve intent inference, coordination, and safe control.

# Questions to refine the research direction

- ▶ Which physical co-manipulation task is the most useful initial test case?
- ▶ What sensing and robotic platforms are currently available?
- ▶ Should the first contribution emphasize human intent inference, interactive response prediction, or planning and control?
- ▶ Which components of RISE Lab's existing game-theoretic framework can be extended directly to physical coupling?
- ▶ What level of cross-task generalization would be most valuable for the current project?

## Discussion objective

Identify a focused first study that contributes to the larger goal of a generalizable and interactive human model.